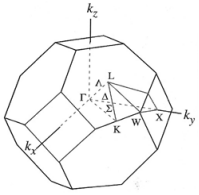
上海科技大学
ShanghaiTech University

Fundamentals of Semiconductor Devices

Baile Chen

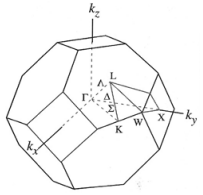
School of Information Science and Technology

Shanghai Tech University



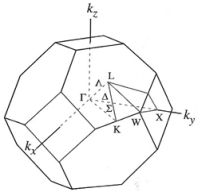
Outline

- Course Information
  - Textbooks
  - Office Hours and TA
  - Topics to be covered
  - Evaluation
  - Homework
  - Exam
  - Project
- Motivation-why study semiconductor devices?



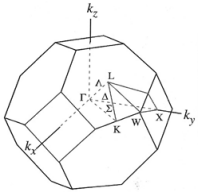
Course Information

- Textbooks
  - Semiconductor Physics And Devices Donald Neamen
  - Solid State Electronic Devices Ben Streetman, Sanjay Banerjee
- Reference Books
  - Physics of Semiconductor Devices
  - Semiconductor Device Fundamentals
  - Advanced Semiconductor Fundamentals
  - Fundamentals of III-V Devices: HBTs, MESFETs, and HFETs/HEMTs



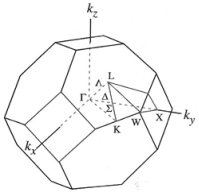
Course Information

- Office Hours
  - Schedule as needed
  - Email: chenbl@shanghaitech.edu.cn
- TA
  - 顾丁然



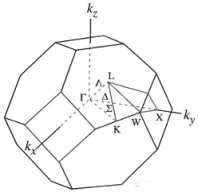
Course Information

- Topics to be covered
  - ➢Semiconductor materials (6-7 weeks)
    - ✓Crystal and Energy band
    - ✓Carrier distribution and Density of State
    - ✓Doping
    - ✓Carrier transport, generation and recombination
  - ➢Semiconductor devices (6-7 weeks)
    - ✓PN junction and PIN junction
    - ✓Photodetectors and LED
    - ✓BJT
    - ✓MOS capacitor
    - ✓MOSFET
  - ➢Class projects (2 weeks)



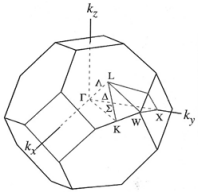
Evaluation

- Homework: 15%
- Midterm: 25% (one A4 page)
- Class Project: 20% (or more)
- Class Participation: 10%
- Final: 30% (one A4 page)



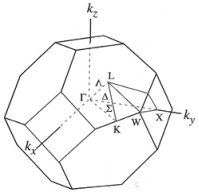
Homework

- Plan to have around 9 assignments
- Each assignment needs to be completed within one week from the date of posting it



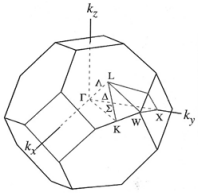
Exam

- Open book with one A4 paper for midterm exam
- Open book with one A4 paper for final exam
- A zero grade will be given for any miss exam

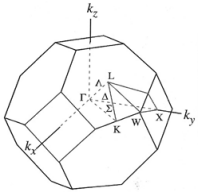


Project

- Three students per group
- Two formats:
  - Paper review
  - Modeling with TCAD or other software (Bonus points)
- Paper review: Students will be asked to research **ONE topic of current interest** to the semiconductor community, to prepare a 15-20 minute presentation on their topic, and to also submit a **8 to 10 page** term paper on the same topic with IEEE paper format. You really need to read **many papers (20+)**, and understand very well about what happened in that area.
- Modeling: review the area first, then repeat (extend if possible) the results of some theoretical data in the papers.
- Everything in English
- Write proposal before 10th week
  - Tell me the area you want to work on

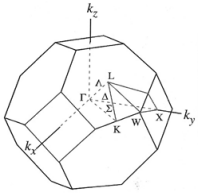


Why do we need this course?



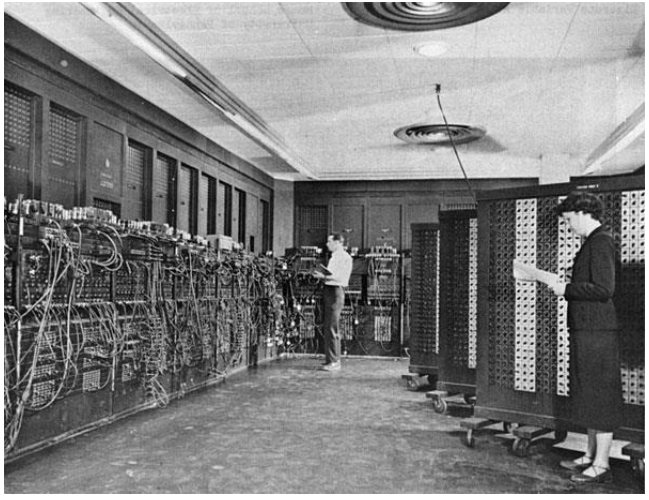
Relation to Other Courses

- System Level:
  - RF design
  - VLSI design
- Application specific device operation:
  - Optoelectronic Devices
  - Advanced Semiconductor Device
  - Bio-devices
- Physical principle of device operation
  - **Fundamentals of Semiconductor Devices**
  - Electromagnetism
- Foundation:
  - Semiconductor Physics
  - Quantum Mechanics
  - Statistical Mechanics
  - Solid State Physics

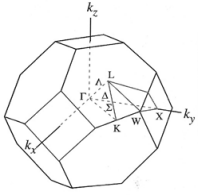


The First Electronic Computer (ENIAC)

- Electronic Numerical Integrator And Calculator
- 50 tons, including ~17,468 vacuum tubes, relays
- Uses 200kW of electricity and cost $500K USD
- It failed every five days

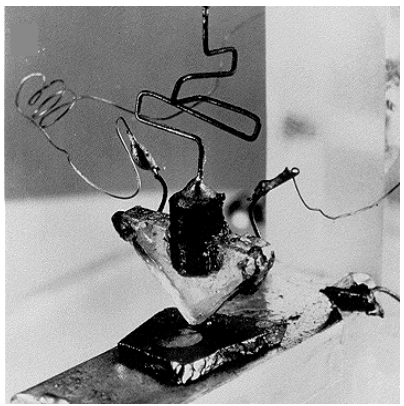


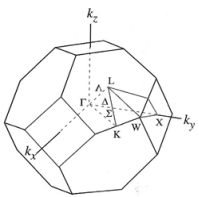
http://www.computerhope.com/jargon/e/eniac.htm



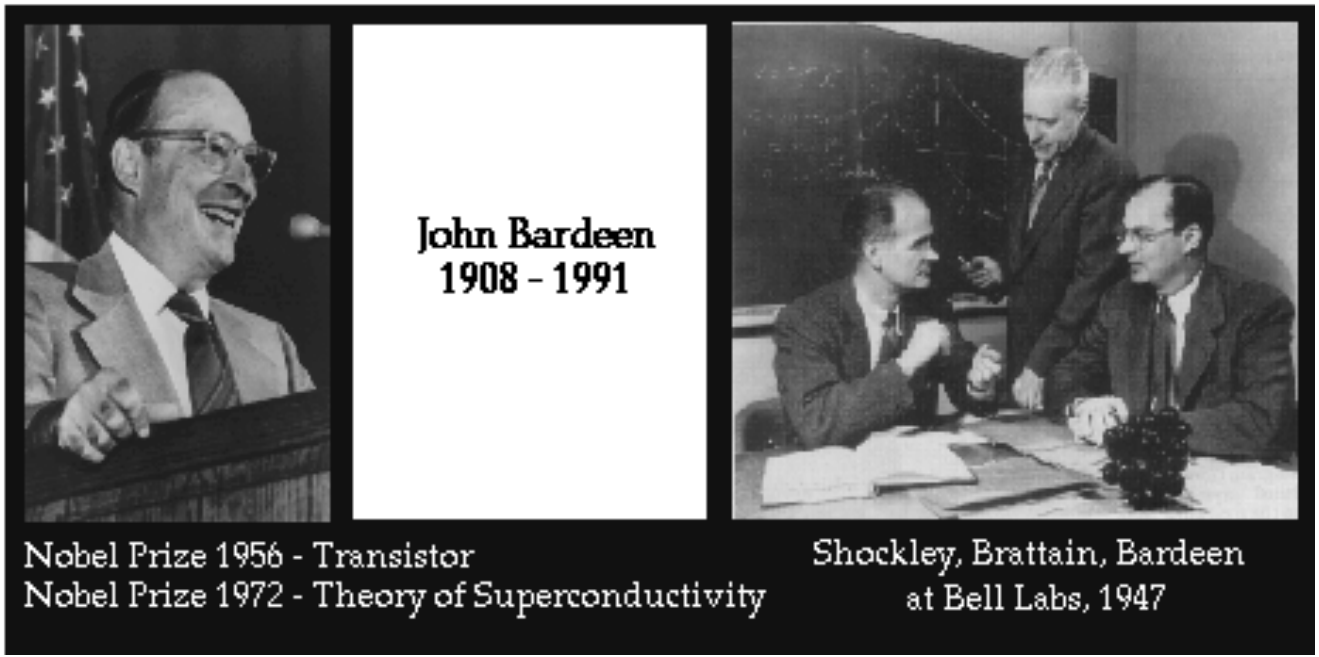
Modern Age

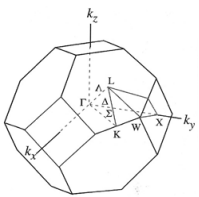
- The first semiconductor transistor
- AT&T Bell Labs, Dec 1947
- J. Bardeen, W. Brattain, W. Shockley
- Germanium base, gold foil contacts





John Bardeen



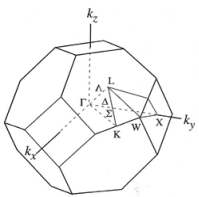


First Transistor Radio Regency TR-1

- Built with four n-p-n transistors and one diode.

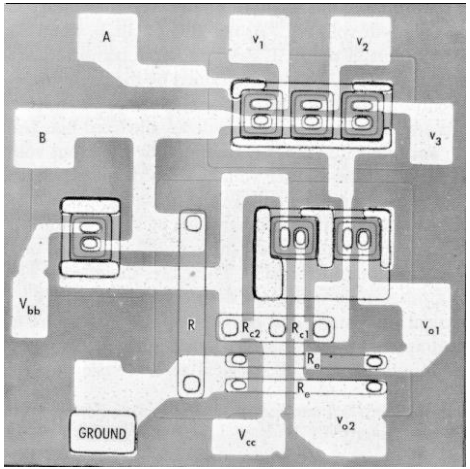


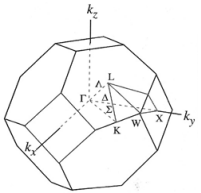
From wiki



Integrated Circuits

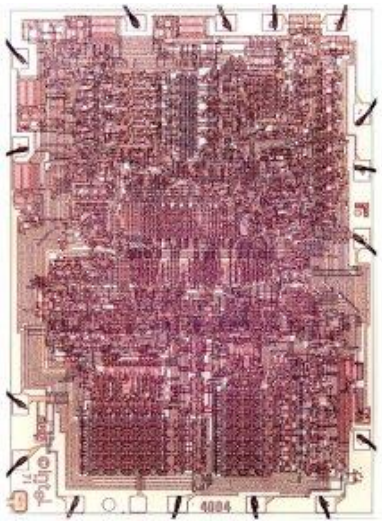
- Fabricate all transistors and metal interconnects on the same piece of silicon
  - ➢Jack Kilby, Nobel prize 2000
  - ➢Robert Noyce 1961, co-founder of Fairchild, then Intel





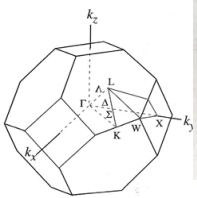
The First Microprocessor, Intel 4004 (1971)

- 2250 transistors, 740 kHz operation
- Comparable computational power with ENIAC
- Built on 2” and then 3” wafers (vs 15” today)
- 10μm line width (vs. 10nm today)
- 4 bit bus width
- Used in Busicom Calculator



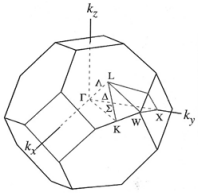




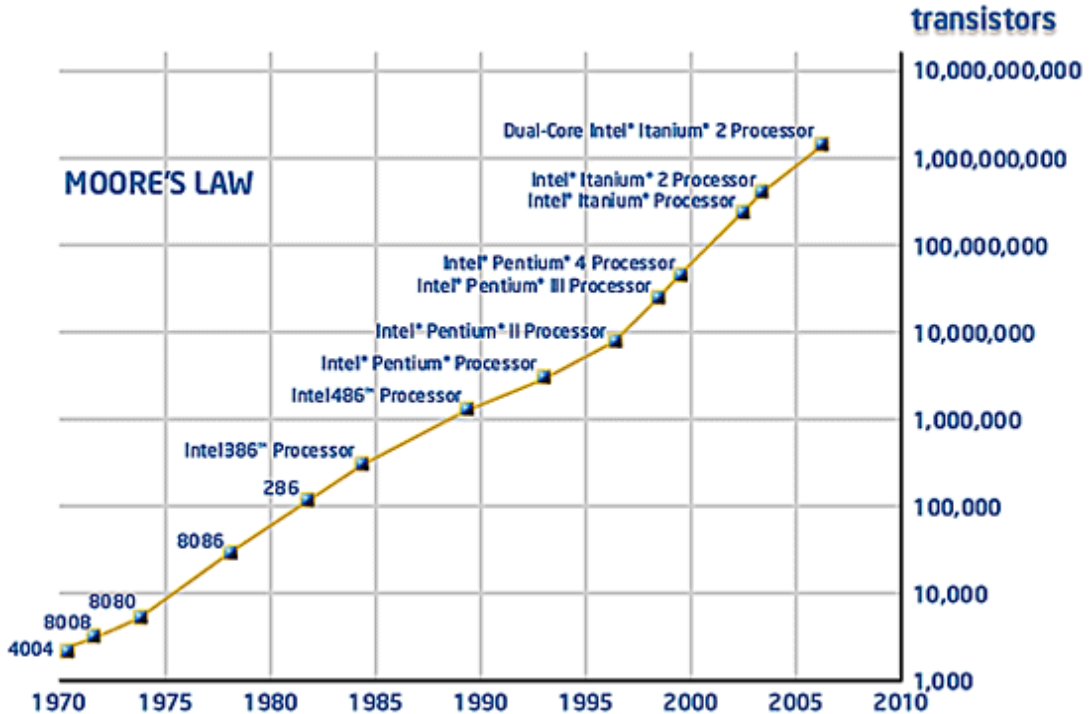


Next

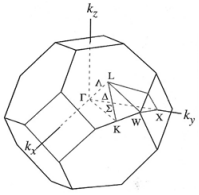
- Followed by 8008 (8-bit), 8080, 8086
- Then 80286, 80386, 80486 = i486 (1989, 0.8 μm lines)
- Pentium, II, III, Itanium, IV, Celeron, Dual-Core, Core 2…



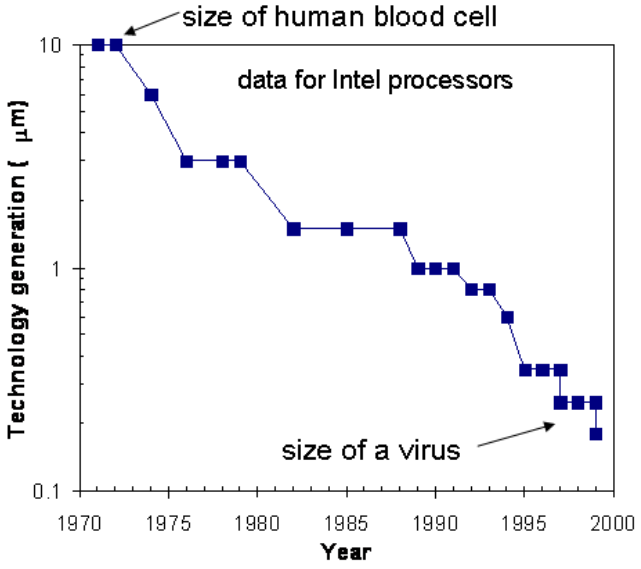
Gordon Moore's "Law"

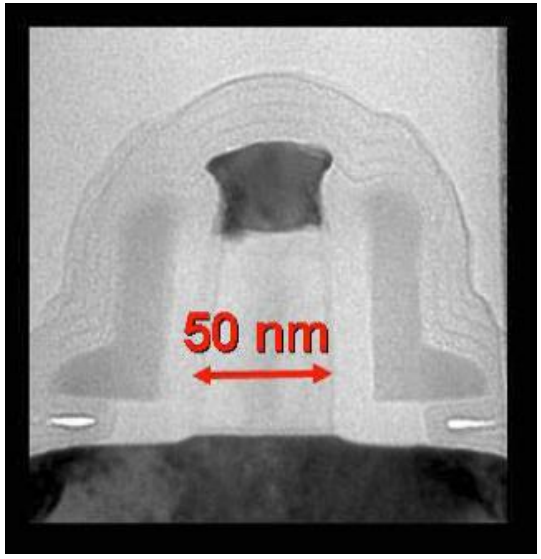


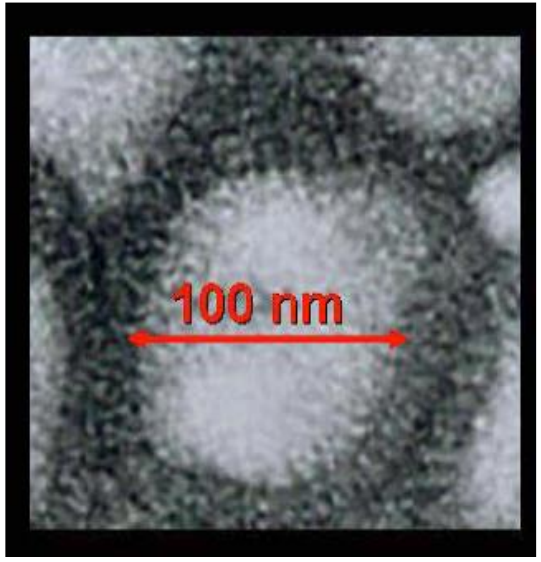
http://www.intel.com



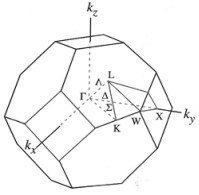
Transistor Size Scaling







Influenza virus



Sources: NSF, Intel

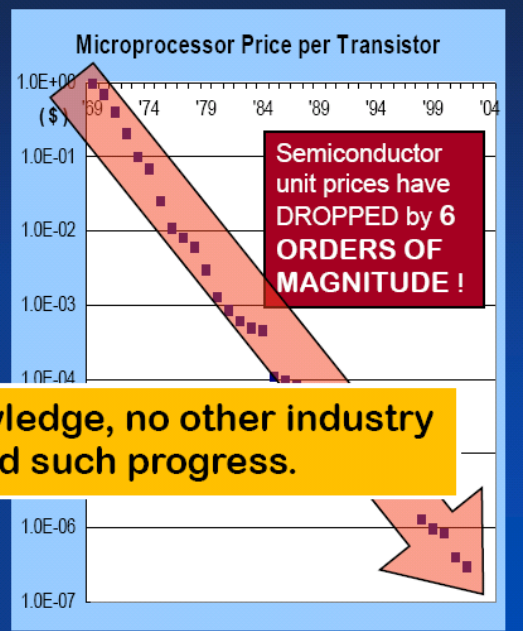
Economic Incentives

Historic Perspective

Over the past 35 years, prices for almost everything have CLIMBED by 5 -10 x.

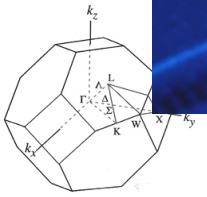
	1970	2008
Avg. Income	$9,300	$36,000
New House	$23,400	$215,000
New Car	$3,900	$26,000
Gallon Gas	$0	
Eggs, dozen	$0	
Postage Stamp	$0.04	$0.41
Burger	$0.20	$0.99

But NOT for semiconductors !

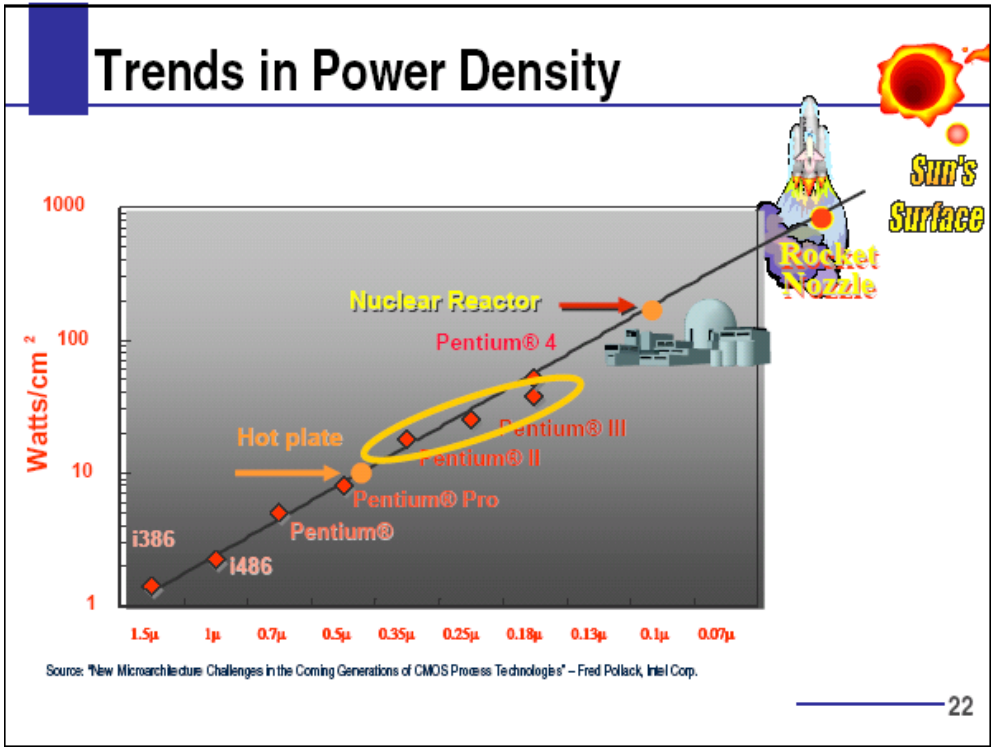


To our knowledge, no other industry has achieved such progress.

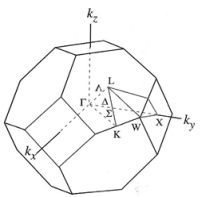
From Ralph Cavin,
NSF-Grantees' Meeting, Dec 3 2008



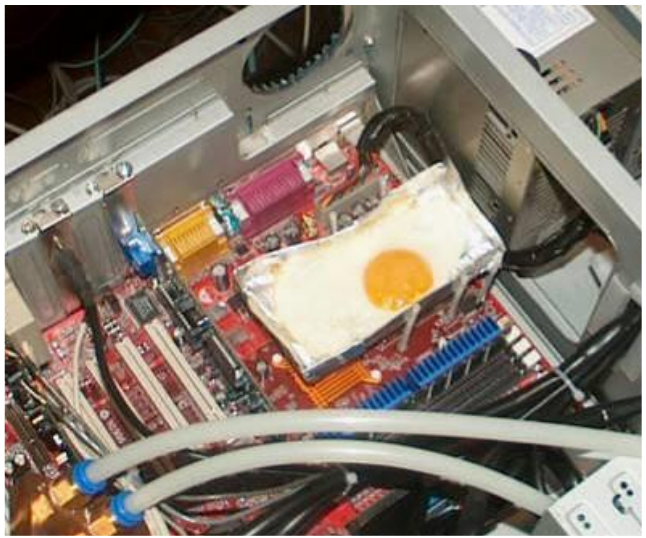
A crisis of epic proportions: Power dissipation !

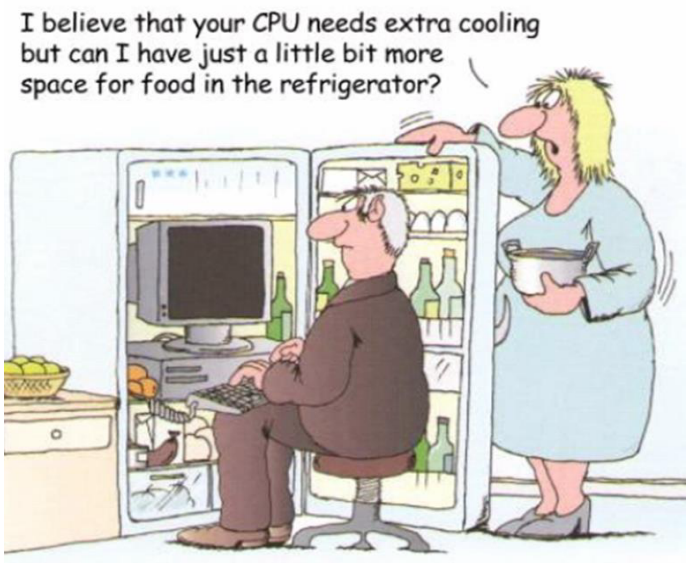


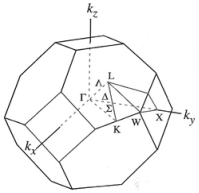
New physics needed – new kinds of computation



Some Negative Side-effects of Scaling:







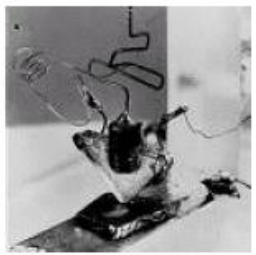
We stand at a threshold in electronics !!

Vacuum Tubes



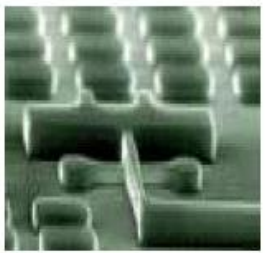
1906-1950s

Bipolar



1947-1980s

MOSFET



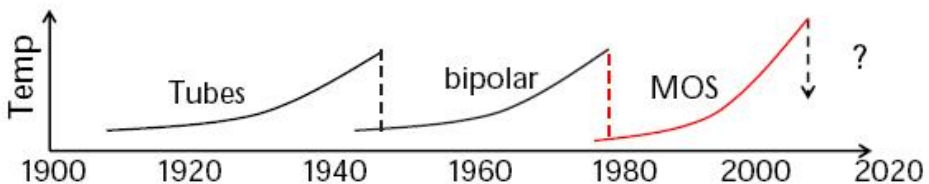
1960-until now

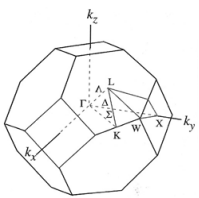
Now ??

Spintronics

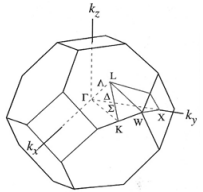
Bio Sensors

Displays ….

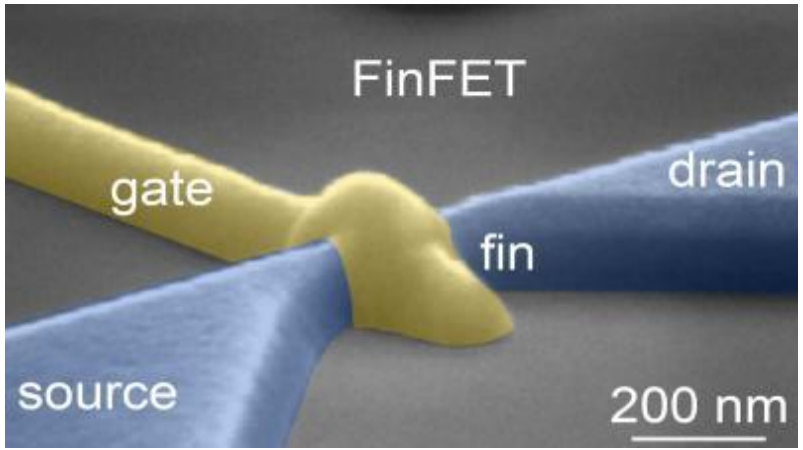


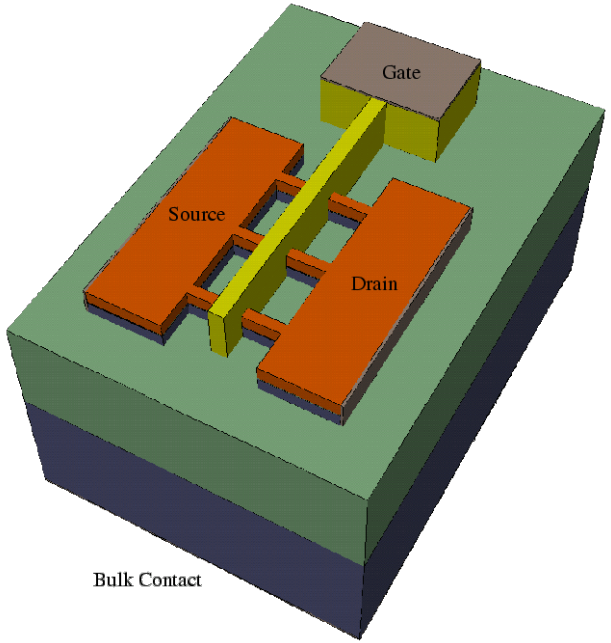


How can we push technology forward?

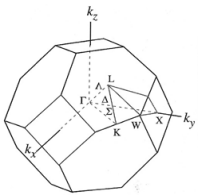


Better Design/Architecture



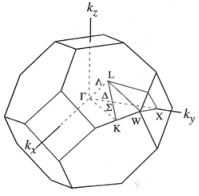


Multiple Gates for superior field control



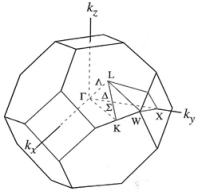
Chenming Hu



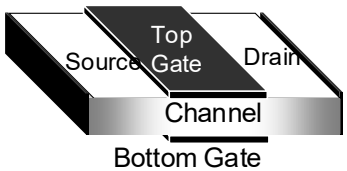


Chenming Hu in Shanghai Tech

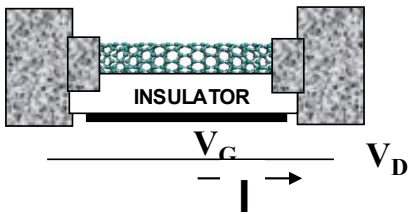




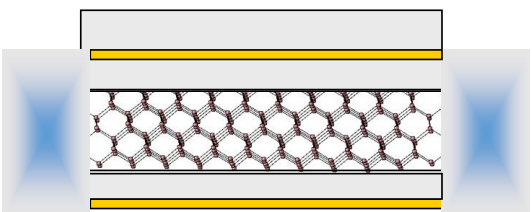
Better Materials?



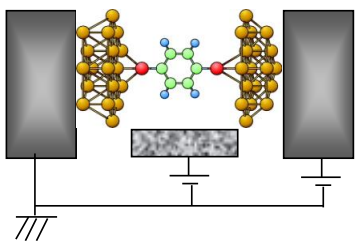
Strained Si, SiGe



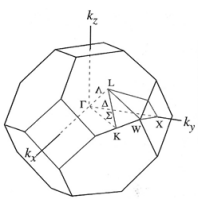
Carbon Nanotubes



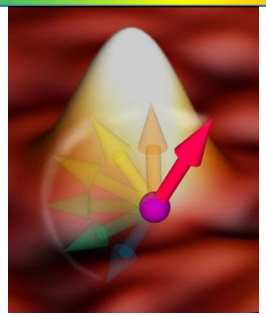
Silicon Nanowires



Organic Molecules



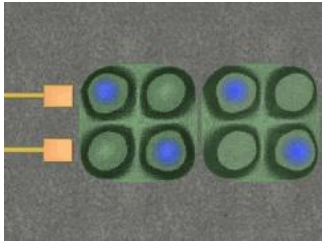
New Principles?



SPINTRONICS

Encode bits in electron's Spin -- Computation by rotating spins

GMR (Nobel, 2007)
MRAMs
STT-RAMs



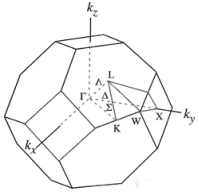
QUANTUM CELLULAR AUTOMATA

Encode bits in quantum dot dipoles



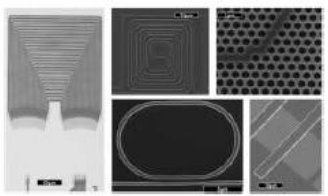
BIO-INSPIRED COMPUTING

Exploit 3-D architecture and massive parallelism

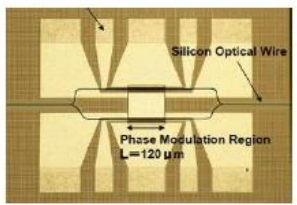


Silicon Photonics

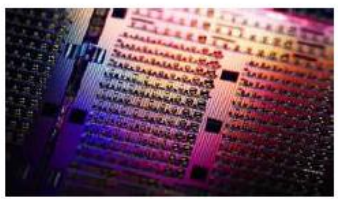
- Multi-core chip with seriously problem of Cu interconnectors
- Si Photonics is the best solution so far for the connectors between the core in the silicon chips



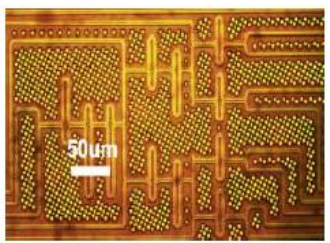
Various Waveguides



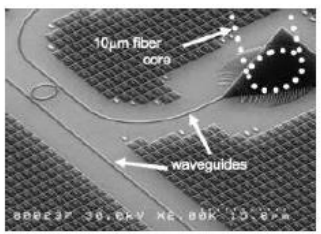
25G Modulator (NEC)



340G APD (Intel)



8 channel WDM (IBM)



-0.8dB Vertical Coupler(Helios)

What is missing in silicon photonics?
Electrically-pumped silicon Lasers

